AI-Powered Event Management System using n8n and Generative AI

Comprehensive Project Report

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Acknowledgement

I sincerely thank my faculty members and mentors for their constant guidance and encouragement during this project. Throughout the development process, I gained valuable experience in workflow automation, API integration, debugging and AI-assisted software development. The motivation behind this project came from observing how much time student organizers spend handling repetitive administrative work before, during and after technical events. Tasks such as verifying registrations, sending confirmation emails, generating participant IDs, tracking attendance and reviewing feedback often consume more time than the actual planning of the event. This project was undertaken with the aim of reducing that repetitive workload through intelligent automation.

Abstract

This project presents an AI-powered Event Management System developed using n8n, Google Forms, Google Sheets, Gmail, Google Drive, QuickChart API and Generative AI. The solution automates participant registration, QR code generation, attendance tracking, reminder emails, event notifications and AI-based feedback analysis. Instead of treating each task as an isolated process, the system integrates five workflows into one ecosystem where information flows through Google Sheets as the central database. The result is a scalable platform that minimizes manual effort, improves communication accuracy and provides organizers with actionable insights after the event.

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Introduction

Event management is a multidisciplinary activity that involves registration management, communication, attendance tracking, coordination among volunteers and post-event evaluation. As the number of participants increases, manual handling becomes increasingly difficult. Automation helps reduce repetitive work while ensuring consistent communication and accurate record keeping. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Problem Analysis

Traditional event management depends heavily on spreadsheets and manual communication. Organizers repeatedly copy participant information, prepare attendance lists, send reminder emails and analyze feedback manually. These repetitive tasks increase the probability of human error and consume valuable planning time. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Existing System

Most student events rely on Google Forms and manually maintained spreadsheets. Confirmation emails are often sent individually, attendance is recorded manually and feedback is reviewed one response at a time. Such systems work for small events but do not scale effectively. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Proposed System

The proposed solution integrates workflow automation and AI into a single platform. Google Forms collect participant information, Google Sheets store data, Gmail automates communication, Google Drive stores QR codes and Gemini AI generates personalized content and analyzes feedback. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Functional Requirements

The system must register participants, generate unique IDs, create QR codes, send confirmation emails, update attendance, deliver reminder emails, generate announcements and analyze participant feedback. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Non-Functional Requirements

The system should be reliable, scalable, easy to maintain, secure, user-friendly and capable of handling multiple participants without significant performance degradation. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Multi-Agent Architecture

The solution is organized as specialized logical agents. The Registration Agent handles participant onboarding, the Attendance Agent validates QR codes, the Reminder Agent manages scheduled communication, the Notification Agent creates AI-generated announcements and the Feedback Agent summarizes participant responses. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Technologies Used

The implementation uses n8n for orchestration, Google Forms for input collection, Google Sheets as the database, Gmail for communication, Google Drive for storage, QuickChart API for QR generation, Gemini AI for natural language generation and JavaScript nodes for custom processing. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Workflow 1 – Registration & Participant Management

Trigger: Google Sheets Trigger. Steps: Generate Registration ID → Gemini → QR Generation → Google Drive → Gmail → Update Sheet.

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Workflow 2 – QR Check-in System

Trigger: Webhook. Steps: Verify Registration ID → Search Sheet → IF Validation → Update Attendance → Respond to Webhook.

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Workflow 3 – AI Reminder Email Automation

Trigger: Schedule Trigger. Steps: Read participants → Gemini → Merge → Gmail.

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Workflow 4 – AI Notification Automation

Trigger: Schedule Trigger. Steps: Read data → JavaScript → Gemini → Gmail.

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Workflow 5 – AI Feedback Analyzer

Trigger: Google Sheets Trigger. Steps: Gemini Analysis → Merge → Update Feedback Sheet.

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Testing

Each workflow was executed multiple times with different inputs. Authentication, merge operations, conditional logic, webhook execution and AI outputs were verified individually before testing the integrated system. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Results

The platform successfully automated repetitive administrative activities. Confirmation emails, QR generation, attendance updates and AI analysis worked as expected, reducing manual effort and improving consistency. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Challenges

OAuth authentication, expression mapping, merge node configuration, webhook testing and structured AI outputs required iterative debugging during development. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Future Scope

Future enhancements include dashboards, certificate generation, WhatsApp integration, mobile applications, payment gateway integration, multilingual support and predictive analytics. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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Conclusion

The project demonstrates how workflow automation and Generative AI can simplify end-to-end event management while allowing organizers to focus on delivering a better participant experience. The implementation follows a modular design where every node performs a clearly defined responsibility. Data validation, error handling and reusable workflow design were emphasized to improve maintainability and reliability. This approach allows the same architecture to be reused for workshops, seminars, hackathons and conferences with minimal modifications.

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